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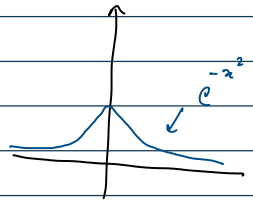
Wednesday, 14 January 2026 10:31 AM

Assumptions : + elastic collisions
+ no external forces ; so density (ρ) independent of position
+ KE is distributed independently and in an equal manner } principle of equipartition
↳ $h(r+s+t) = g(r)\, g(s)\, g(t)$ KE scaling as v^2 or p^2

Maxwell distribution : $g(v_x^2) = \alpha\, e^{-a v_x^2}$

↳ probability

$$\int_{-\infty}^{\infty} f_x(v_x^2) \cdot dv_x = 1$$



Derivations → $|f(\vec{v})| = f(|v|) = f(v^2)$

$\Phi = \frac{\Delta N}{\Delta A\, \Delta t}$ → dependent on probability of particles of v going at an angle

molecular flux

↳ @ equilibrium

$F = \frac{dp}{dt}$ ← change in momentum

kT → energy scale of average KE, or motion of a particle